



Mentora AI

Smart Learning, Bright Future

Mentora

AI Tutor for Personalized Learning

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[GitHub](#)

The Problem with Learning Today

One-Size-Fits-All

Learning is still delivered at a fixed pace, with little room for individual needs.

Tutoring is Expensive

Personalized support remains out of reach for most students, ranging from 35-100\$ per hour.

Passive Online Learning

Most digital content informs, but does not actively build understanding.

No Real Feedback & Teaching

Students need immediate, adaptive guidance — not just answers.



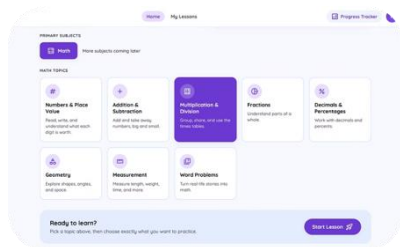
Meet Mentora

An AI-powered tutor that explains, questions, and adapts – in real conversation.



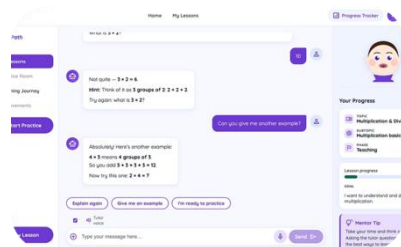
Mentora replaces passive study with a dynamic, voice-driven tutoring loop – personalized to every student, available 24/7.

Product Walkthrough



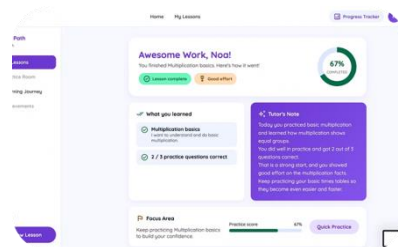
Subject & Topic Selection

Browse subjects and choose a lesson topic from a clean, card-based dashboard.



Conversational Tutor

The AI tutor explains concepts, asks follow-up questions, and evaluates answers through voice.



Progress Tracking

Visual progress cards and completion stats help students stay motivated and on track.

System Architecture

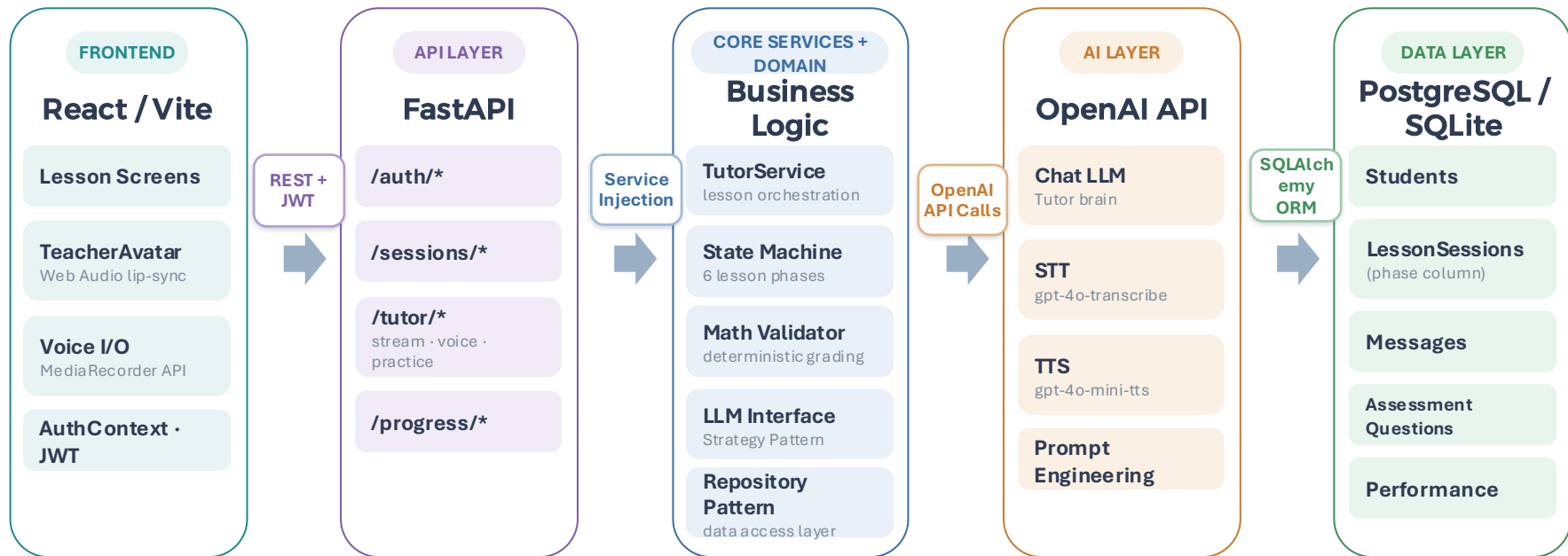
React

FastAPI

OpenAI API

SQLAlchemy

High-Level System Architecture



Streaming Voice Loop

Student speaks → MediaRecorder → STT → Chat LLM → TTS → 32KB audio chunks → Avatar lip-sync (Web Audio AnalyserNode)

State-Based Lesson Flow

Teaching → Pre-Practice Example → Practice → Practice Summary → Summary → Completed · Each phase is an isolated State class

Deterministic Math Grading

6-tier validator (Fraction · SymPy · Equation · Percentage · String · LLM fallback) · LLM never owns is_correct · prevents hallucinated grades

Progress Tracking

Every practice set persisted with set_number, difficulty, is_correct · aggregated into mastery map per topic/subtopic · success distribution dashboard

Challenges & Decisions

Key engineering trade-offs and how we resolved them

LLM Reliability

LLM grading was non-deterministic for math, so we built a deterministic verification layer using SymPy. The LLM never decides `is_correct` — it only generates feedback.

TTS Latency

Text-to-speech is streamed in 32KB audio chunks so playback begins before full synthesis completes, keeping the voice loop responsive.

Prompt Engineering

Structured prompts with strict role separation keep tutoring on-topic and consistent, and prevent the model from leaking answers.

State Management

Each lesson phase is modeled as an isolated State class, making the lesson flow predictable, testable, and easy to extend.

What's Next for Mentora

Smarter Memory

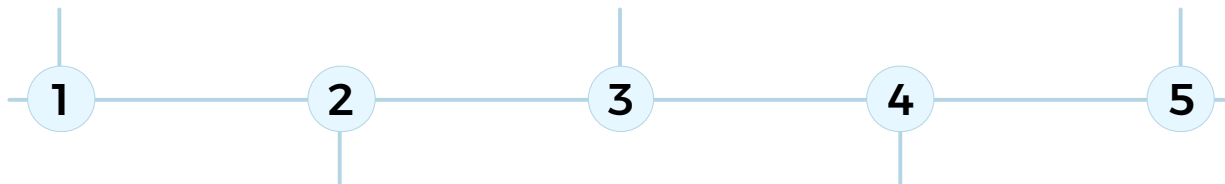
Contextual memory across sessions so the tutor remembers each student's progress and gaps.

Animated Avatar

A fully expressive AI tutor avatar with lip-sync and emotion for a more human learning experience.

More Subjects

Expanding beyond the core subjects — adding Hebrew, English, and Science for broader learning support.



Upload Your Materials

Students upload slides, PDFs, or syllabi — Mentora becomes a tutor for *their* specific course.

Analytics Dashboard

Deep student insights — weak topics, session history, progress curves — for students and educators.

How Mentora Compares

Feature	Mentora	ChatGPT	Khanmigo	SchoolAI	Real Private Tutor
Lesson plan management & curriculum planning	✓	✗	✓	✓	✓
Audio & video interaction	✓	✗	✓	✗	✓
Progress tracking & analytics	✓	✗	✓	✓	✓
Personalized adaptive instruction	✓	✗	✓	✓	✓
Document upload & homework assistance	✓	✓	✓	✓	✓
Parental controls & progress reporting	✓	✗	✗	✓	✓
Hebrew & multilingual interface	✓	✓	✗	✗	✓
Availability	✓	✓	✓	✓	✗
Pricing	Low	Low	Low	Medium	High

Our Business Model

Free to start, simple to upgrade — with plans for students, families and schools.



Built for ages 6–15

FREE

Free

\$0 forever

- A limited guided lessons per day
- Video tutoring
- Basic progress tracking

MOST POPULAR

Premium

\$10 / month

- Unlimited tutoring sessions
- Unlimited file uploads
- Detailed progress reports

FOR FAMILIES

Family

\$10 + 5\$ for each added child

- Everything in Premium
- Parent / teacher dashboard
- Tiered pricing as you add kids



Schools & Institutions

Annual site license — every student gets full access.

The real buyer is the educator. Family & school tiers turn one happy learner into many seats.

Mentora: A Tutor for Every Student



Truly Personalized

An AI tutor that explains, questions and adapts to every student – 24/7.



Full Voice Loop

Speak or type – transcribe, reason and speak back in seconds. Working today.



Built to Scale

Clean React + FastAPI + GPT stack – ready for new subjects and users.

Thank you!
Noa Levy & Ben Garusi ·
Questions?

